

Computation for ϕ^3 Theory

Follows Srednicki and Peskin & Schroeder — counter-terms, partition functional, and one-loop diagrams in ϕ^3 .

Yucheng (Luca) Jin

Quantum Field Theory

1. Computation for ϕ^3 Theory

All of the calculation below is based on Srednicki, and Peskin and Schroeder, so nothing here is original at all.

Here, we will use the metric signature $(-, +, +, +)$. The Lagrangian of ϕ^3 is given by

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 + \frac{1}{3!}g\phi^3. \quad (1.1)$$

To make the theory satisfying

$$\langle k | \phi(x) | \Omega \rangle = e^{-ik \cdot x}, \quad \langle \Omega | \phi(x) | \Omega \rangle = 0, \quad (1.2)$$

we make field redefinitions so that the Lagrangian becomes

$$\mathcal{L} = -\frac{1}{2}Z_\phi\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}Z_m m^2\phi^2 + \frac{1}{3!}Z_g g\phi^3 + Y\phi = \mathcal{L}_0 + \mathcal{L}_1, \quad (1.3)$$

where

$$\mathcal{L}_0 = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2, \quad (1.4)$$

and

$$\mathcal{L}_1 = -\frac{1}{2}(Z_\phi - 1)\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}(Z_m - 1)m^2\phi^2 + \frac{1}{3!}Z_g g\phi^3 + Y\phi. \quad (1.5)$$

The partition functional can be written as

$$Z = \int \mathcal{D}\phi(x) e^{i \int d^d x (\mathcal{L}_0 + J\phi + \mathcal{L}_1)} = e^{i \int d^d x \mathcal{L}_1(-i\delta/\delta J(x))} \int \mathcal{D}\phi(x) e^{i \int d^d x (\mathcal{L}_0 + J\phi)} = e^{i \int d^d x \mathcal{L}_1(-i\delta/\delta J(x))} Z_0, \quad (1.6)$$

where

$$Z_0 = \int \mathcal{D}\phi(x) e^{i \int d^d x (\mathcal{L}_0 + J\phi)}. \quad (1.7)$$

Making the field redefinition

$$\chi(x) = \phi(x) - iD_F(x-y)J(y), \quad (1.8)$$

where

$$(\partial^2 - m^2)D_F(x-y) = i\delta^d(x-y). \quad (1.9)$$

Note that since the above is just a shift of field, the measure is unchanged $\mathcal{D}\phi = \mathcal{D}\chi$. Then, the integrand of Z_0 becomes

$$\exp \left[i \int d^d x \left(\frac{1}{2} \chi(x) (\partial^2 - m^2) \chi(x) + \frac{i}{2} \int d^d y J(y) D_F(x-y) J(y) \right) \right], \quad (1.10)$$

and hence

$$Z_0 = \mathcal{N} \exp \left[-\frac{1}{2} \int d^d x \int d^d y J(x) D_F(x-y) J(y) \right], \quad (1.11)$$

where $\mathcal{N}$ is formally infinite, but since it is not a measurable quantity, it does not boarder us. Now the full partition functional becomes

$$Z = e^{i \int d^d x \mathcal{L}_1(-i\delta/\delta J(x))} \exp \left[-\frac{1}{2} \int d^d x \int d^d y J(x) D_F(x-y) J(y) \right] \quad (1.12)$$

up to some irrelevant constant.

Now, we are interested in finding the exact two-point correlation function to one-loop order. That is

$$\langle \Omega | T \phi(x_1) \phi(x_2) | \Omega \rangle = (-i)^2 \frac{\delta^2}{\delta J(x_1) \delta J(x_2)} \ln Z[J] \Big|_{J=0} = i(-i)^2 \frac{\delta^2}{\delta J(x_1) \delta J(x_2)} W[J] \Big|_{J=0}, \quad (1.13)$$

where $W[J]$ is the connected diagram generating functional

$$Z[J] \equiv e^{iW[J]}. \quad (1.14)$$

In practice, one can simply write down the Feynman rules and compute the correlation functions. However, for our purpose, we shall compute it directly from functional derivatives. We shall first compute $Z[J]$:

$$\begin{aligned} Z[J] &= \left[1 + i \int d^d x \frac{1}{2} \left(-i \frac{\delta}{\delta J(x)} \right) (A\partial_x^2 - Bm^2) \left(-i \frac{\delta}{\delta J(x)} \right) + \dots \right] Z_0 \\ &= Z_0 - \frac{i}{2} \int d^d x \left(-i \frac{\delta}{\delta J(x)} \right) (A\partial_x^2 - Bm^2) \left(- \int d^d y J(y) D_F(x-y) \right) Z_0 + \dots \\ &= Z_0 - \frac{i}{2} \int d^d x \left(\int d^d z J(z) D_F(x-z) \right) (A\partial_x^2 - Bm^2) \left(\int d^d y J(y) D_F(x-y) \right) Z_0 + \dots, \end{aligned} \quad (1.15)$$

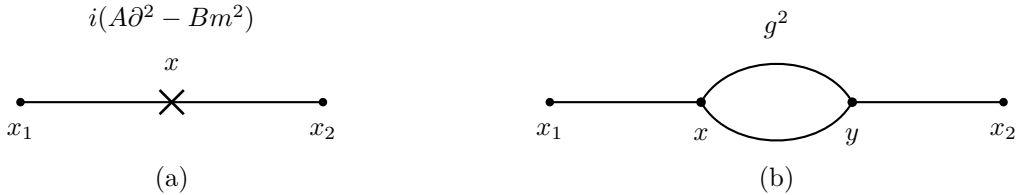
where we have focused on the connected two-point term since we will eventually take $J = 0$ after the functional derivative. However, one should not forget the other term

$$\begin{aligned} &\frac{i^2}{(3!)^2} g^2 \int d^d x \int d^d y \left(-i \frac{\delta}{\delta J(x)} \right)^3 \left(-i \frac{\delta}{\delta J(y)} \right)^3 Z_0 \\ &= -\frac{g^2}{2} \int d^d x \int d^d y J(x) D_F(x-y)^2 J(y) Z_0 + \dots \end{aligned} \quad (1.16)$$

Now, we can compute the two-point correlation function; that is

$$\begin{aligned} D(x_1 - x_2) &= \langle \Omega | T \phi(x_1) \phi(x_2) | \Omega \rangle = - \frac{\delta^2}{\delta J(x_1) \delta J(x_2)} Z[J] \Big|_{J=0} \\ &= D_F(x_1 - x_2) + i \int d^d x D_F(x - x_1) (A\partial_x^2 - Bm^2) D_F(x - x_2) \\ &\quad - \frac{g^2}{2} \int d^d x \int d^d y D_F(x_1 - x) D_F(x - y)^2 D_F(y - x_2) + \mathcal{O}(g^4), \end{aligned} \quad (1.17)$$

where we have been quite sloppy with the overall constant throughout. These are exactly the diagrams one would expect from Feynman rules



In momentum space, we have

$$\tilde{D}(k^2) = \tilde{D}(k^2) + \tilde{D}_F(k^2) i \Pi(k^2) \tilde{D}_F(k^2) + \mathcal{O}(g^4), \quad (1.18)$$

where

$$i \Pi(k^2) = -i(Ak^2 + Bm^2) - \frac{g^2}{2} \int \frac{d^d l}{(2\pi)^d} \frac{i}{l^2 + m^2 - i\epsilon} \frac{i}{(k+l)^2 + m^2 - i\epsilon} + \mathcal{O}(g^4). \quad (1.19)$$

Consider the con exact propagator, where $\Pi(k^2)$ is now to all order of g ,

$$\tilde{D}(k^2) = \tilde{D}(k^2) \sum_{n=0}^{\infty} [i \Pi(k^2) \tilde{D}_F(k^2)]^n = \frac{i}{k^2 + m^2 - i\epsilon - \Pi(k^2)}. \quad (1.20)$$

For the particle to have mass m , we need the exact propagator with residue one at $k^2 = -m^2$. Therefore, we require

$$\Pi(-m^2) = \Pi'(-m^2) = 0, \quad (1.21)$$

which helps us fix A and B . Now, we need to compute the integral above.

Now, we shall perform the integral in d dimensions. To do that, we use the Feynman's formula.

Claim:

$$\prod_{j=1}^n \frac{1}{A_j^{\alpha_j}} = \frac{\Gamma(\sum_i \alpha_i)}{\prod_i \Gamma(\alpha_i)} \frac{1}{(n-1)!} \int dF_n \frac{\prod_i x_i^{\alpha_i-1}}{(\sum_i x_i A_i)^{\sum_i \alpha_i}}, \quad (1.22)$$

where

$$dF_n = (n-1)! \int_0^1 dx_1 \cdots \int_0^1 dx_n \delta(1 - \sum_i x_i). \quad (1.23)$$

Proof: Consider

$$\frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} = \int_0^\infty dt_i t_i^{\alpha_i-1} e^{-A_i t_i}. \quad (1.24)$$

Then, we take

$$\prod_{i=1}^n \frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} = \prod_{i=1}^n \int_0^\infty dt_i t_i^{\alpha_i-1} e^{-A_i t_i}. \quad (1.25)$$

We can multiply the right-hand side by

$$1 = \int_0^\infty ds \delta(s - \sum_i t_i), \quad (1.26)$$

which gives

$$\prod_{i=1}^n \frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} = \int_0^\infty ds \delta(s - \sum_i t_i) \prod_{i=1}^n \int_0^\infty dt_i t_i^{\alpha_i-1} e^{-A_i t_i}. \quad (1.27)$$

Now, making the substitution $t_i = s x_i$, we find

$$\begin{aligned} \prod_{i=1}^n \frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} &= \int_0^\infty ds \delta(s - \sum_i x_i) \prod_{i=1}^n \int_0^\infty dx_i s^{\alpha_i} x_i^{\alpha_i-1} e^{-s A_i x_i} \\ &= \frac{1}{(n-1)!} \int dF_n \left(\prod_{i=1}^n x_i^{\alpha_i-1} \right) \int_0^\infty ds s^{\sum_i \alpha_i-1} e^{-\sum_i s A_i x_i} \\ &= \frac{1}{(n-1)!} \int dF_n \left(\prod_{i=1}^n x_i^{\alpha_i-1} \right) \frac{\Gamma(\sum_i \alpha_i)}{(\sum_i A_i x_i)^{\sum_i \alpha_i}}. \end{aligned} \quad (1.28)$$

□

We use

$$\frac{1}{A_1 A_2} = \int_0^1 dx_1 \int_0^1 dx_2 \delta(x_1 + x_2 - 1) \frac{1}{(x_1 A_1 + x_2 A_2)^2} = \int_0^1 dx \frac{1}{[x A_1 + (1-x) A_2]^2}. \quad (1.29)$$

Therefore, we have

$$\begin{aligned} \frac{1}{l^2 + m^2} \frac{1}{(k+l)^2 + m^2} &= \int_0^1 dx [(1-x)(l^2 + m^2) + x((k+l)^2 + m^2)]^{-2} \\ &= \int_0^1 dx [l^2 + 2xkl + m^2 + xk^2]^{-2} \\ &= \int_0^1 dx [(l+xk)^2 + m^2 + x(1-x)k^2]^{-2}, \end{aligned} \quad (1.30)$$

where we have abbreviated $m^2 - i\epsilon$ as m^2 . Now, we write $q \equiv l + xk$ and $D \equiv m^2 + x(1-x)k^2$. So that the integral is now

$$-\frac{g^2}{2} \int \frac{d^d l}{(2\pi)^d} \frac{i}{l^2 + m^2 - i\epsilon} \frac{i}{(k+l)^2 + m^2 - i\epsilon} = \frac{g^2}{2} \int_0^1 dx \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + D)^2}. \quad (1.31)$$

Then, we perform the Wick rotation so that $q^0 = iq_E^0$, $\mathbf{q} = \mathbf{q}_E$, $q^2 = q_E^2$, and $d^d q = id^d q_E$. Therefore, the above integral becomes

$$I(k^2) = i \int_0^1 dx \int \frac{d^d q_E}{(2\pi)^d} \frac{1}{(q_E^2 + D)^2}, \quad (1.32)$$

where

$$i\Pi(k^2) = \frac{g^2}{2} I(k^2) - i(Ak^2 + Bm^2) + \mathcal{O}(g^4). \quad (1.33)$$

With a slight abbreviation of notation, we now write the integral measure as $d^d q_E = q_E^{d-1} dq_E d\Omega_{d-1}$, where q_E is now the length of the vector q_E .

Claim:

$$\Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}. \quad (1.34)$$

Proof: Consider the integral

$$\int_{\mathbb{R}^d} d^d x e^{-x^2} = \int d\Omega_{d-1} \int_0^\infty dr r^{d-1} e^{-r^2}. \quad (1.35)$$

The left-hand side gives

$$\left(\int_{\mathbb{R}} dx e^{-x^2} \right)^d = \pi^{d/2}, \quad (1.36)$$

and the right-hand side gives

$$\Omega_{d-1} \frac{1}{2} \int_0^\infty dr r^{d/2-1} e^{-r} = \Omega_{d-1} \frac{\Gamma(d/2)}{2}. \quad (1.37)$$

□

We shall also prove the general identity, that would turn out to be useful in many cases when dealing with loop corrections.

Claim:

$$\int \frac{d^d q_E}{(2\pi)^d} \frac{(q_E^2)^n}{(q_E^2 + D)^m} = \frac{\Gamma(m-n-d/2)\Gamma(n+d/2)}{(4\pi)^{d/2}\Gamma(m)\Gamma(d/2)} D^{-(m-n-d/2)}. \quad (1.38)$$

Proof:

$$\int \frac{d^d q_E}{(2\pi)^d} \frac{(q_E^2)^n}{(q_E^2 + D)^m} = \frac{2\pi^{d/2}}{\Gamma(d/2)} \frac{1}{(2\pi)^d} \int_0^\infty dq_E \frac{q_E^{2n+d-1}}{(q_E^2 + D)^m}. \quad (1.39)$$

Making the substitution $s^2 = q^2/D + 1$, we find

$$\int \frac{d^d q_E}{(2\pi)^d} \frac{(q_E^2)^n}{(q_E^2 + D)^m} = \frac{2D^{n+d-m-1}}{(4\pi)^{d/2}\Gamma(d/2)} \int_1^\infty ds \frac{(s^2 - 1)^{n+d/2-1}}{s^{2m-1}}. \quad (1.40)$$

Then, we substitute $t = 1/s^2$, we find

$$\begin{aligned} \int \frac{d^d q_E}{(2\pi)^d} \frac{(q_E^2)^n}{(q_E^2 + D)^m} &= \frac{D^{n+d-m}}{(4\pi)^{d/2}\Gamma(d/2)} \int_0^1 dt t^{m-n-d/2-1} (t-1)^{n+d/2-1} \\ &= \frac{D^{n+d-m}}{(4\pi)^{d/2}\Gamma(d/2)} \beta(m-n-d/2, n+d/2-1) \\ &= \frac{D^{n+d-m}}{(4\pi)^{d/2}\Gamma(d/2)} \frac{\Gamma(m-n-d/2)\Gamma(n+d/2)}{\Gamma(m)}. \end{aligned} \quad (1.41)$$

□

For our case, we have $n = 0$ and $m = 2$, and hence

$$I(k^2) = i \int_0^1 dx \frac{\Gamma(2-d/2)}{(4\pi)^{d/2}} D^{-2+d/2}. \quad (1.42)$$

Now let us focus on the ϕ^3 theory in $d = 6$, where $[g] = 0$ and the theory is renormalizable. For general dimensions, $[g] = 3 - d/2 \equiv \varepsilon/2$. We shall redefine the coupling constant to make it dimension less, that is

$$g \rightarrow g\tilde{\mu}^{\varepsilon/2}. \quad (1.43)$$

Now, we find

$$g^2 \tilde{\mu} I(k^2) = ig^2 \tilde{\mu} \int_0^1 dx \frac{\Gamma(\varepsilon/2-1)}{(4\pi)^{3-\varepsilon/2}} D^{-\varepsilon/2+1} = ig^2 \frac{\Gamma(-1+\varepsilon/2)}{(4\pi)^3} \int_0^1 dx D \left(\frac{4\pi \tilde{\mu}^2}{D} \right)^{\varepsilon/2}. \quad (1.44)$$

Using dimensional regularization, we take $\varepsilon \rightarrow 0$ so that

$$\Gamma(-1+\varepsilon/2) \sim -\frac{2}{\varepsilon} + \gamma - 1 + \dots, \quad (1.45)$$

and

$$\left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\varepsilon/2} \sim 1 + \frac{\varepsilon}{2} \ln\left(\frac{4\pi\tilde{\mu}^2}{D}\right) + \dots \quad (1.46)$$

Then,

$$\Gamma(-1 + \varepsilon/2) \left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\varepsilon/2} \sim -\frac{2}{\varepsilon} + \gamma - 1 - \ln\left(\frac{4\pi\tilde{\mu}^2}{D}\right) + \mathcal{O}(\varepsilon). \quad (1.47)$$

Defining

$$\mu = \sqrt{4\pi}e^{-\gamma/2}\tilde{\mu}, \quad \alpha = \frac{g^2}{(4\pi)^3} \quad (1.48)$$

we find

$$g^2\tilde{\mu}I(k^2) \sim i\alpha \int_0^1 dx D \left[-\frac{2}{\varepsilon} - 1 + \ln\left(\frac{m^2}{\mu^2}\right) - \ln\left(\frac{m^2}{D}\right) + \mathcal{O}(\varepsilon) \right]. \quad (1.49)$$

Now, we evaluate

$$\int_0^1 dx D = \int_0^1 dx (m^2 + x(1-x)k^2) = m^2 + k^2/6. \quad (1.50)$$

Therefore, we find

$$\begin{aligned} \Pi(k^2) &\sim \frac{\alpha}{2} \left[-(2/\varepsilon + 1 + 2\ln(\mu/m))(m^2 + k^2/6) + \int_0^1 dx D \ln(D/m^2) \right] - (Ak^2 + Bm^2) + \mathcal{O}(\alpha^2) \\ &= \frac{\alpha}{2} \int_0^1 dx D \ln(D/m^2) - \left[\frac{1}{6}\alpha(\varepsilon + \ln(\mu/m) + 1/2) + A \right] k^2 \\ &\quad - [\alpha(\varepsilon + \ln(\mu/m) + 1/2) + A] m^2 + \mathcal{O}(\alpha^2). \end{aligned} \quad (1.51)$$

To further simplify the expression, we shall let A and B to take the form

$$A = -\frac{1}{6}\alpha(\varepsilon + \ln(\mu/m) + 1/2) + \alpha\kappa_A - \mathcal{O}(\alpha^2), \quad (1.52)$$

$$B = -\alpha(\varepsilon + \ln(\mu/m) + 1/2) - \alpha\kappa_B + \mathcal{O}(\alpha^2). \quad (1.53)$$

Then, we find

$$\Pi(k^2) \sim \frac{\alpha}{2} \int_0^1 dx D \ln(D/m^2) - \alpha \left(\frac{1}{6}\kappa_A k^2 + \kappa_B m^2 \right) + \mathcal{O}(\alpha^2). \quad (1.54)$$

Now, we impose the renormalization condition $\Pi(-m^2) = \Pi'(-m^2) = 0$ to fix κ_A and κ_B . To do that, we note that $\Pi(-k^2)$ must take the form

$$\Pi(-k^2) \sim \frac{\alpha}{2} \int_0^1 dx D \ln(D/D_0) + c(k^2 + m^2) + \mathcal{O}(\alpha^2), \quad (1.55)$$

where $c \in \mathbb{R}$ and

$$D_0 = D|_{k^2=-m^2} = [1 - x(1-x)]m^2. \quad (1.56)$$

Then, we differentiate with respect to k^2 to find

$$\Pi'(-k^2) \sim \frac{\alpha}{2} \int_0^1 dx [x(1-x) \ln(D/D_0) + D_0 x(1-x)] + c + \mathcal{O}(\alpha^2). \quad (1.57)$$

Therefore,

$$\Pi'(-m^2) = 0 \implies c = -\frac{\alpha}{12}. \quad (1.58)$$

Finally we obtain the 1-loop correction for the two-point correlation function, which is

$$\Pi(-k^2) \sim \frac{\alpha}{2} \int_0^1 dx D \ln(D/D_0) - \frac{\alpha}{12}(k^2 + m^2) + \mathcal{O}(\alpha^2). \quad (1.59)$$

We might as well right down the asymptotic value for κ_A and κ_B . That is

$$\kappa_A = \frac{1}{4}, \quad \kappa_B = \frac{1}{24}. \quad (1.60)$$

Note that the integral in our final expression is real if $D > 0$. That is $x(1-x)k^2 > -m^2$, and since $x(1-x) \in [0, 1/4]$ we must have $k^2 > -4m^2$ for real $\Pi(k^2)$. Then, one might be tempting to ask: what happens if $k^2 < -4m^2$, and $\text{Im } \Pi(k^2) \neq 0$? To understand what is happening, let us write the one-loop correction in Lehmann-Källén form,

$$\tilde{D}(k^2) = \frac{i}{k^2 + m^2 - i\epsilon} + \int_{4m^2}^{\infty} ds \rho(s) \frac{i}{k^2 + s - i\epsilon} = \frac{i}{k^2 + m^2 - i\epsilon - \Pi(k^2)}. \quad (1.61)$$

Using the identity

$$\frac{1}{x - i\epsilon} = P \frac{1}{x} + i\pi\delta(x), \quad (1.62)$$

we find

$$\text{Im}(-i\tilde{D}(k^2)) = \pi\delta(k^2 + m^2) + \pi\rho(-k^2) = \pi\delta(k^2 + m^2 - \Pi(k^2)), \quad (1.63)$$

where $k^2 > -4m^2$. Therefore, for $k^2 > -4m^2$, $\rho(-k^2) = \rho(s) = 0$. However, for $k^2 < -4m^2$, $\text{Im } \Pi(k^2) \neq 0$. Hence, one would find

$$\pi\rho(s) = \frac{\text{Im } \Pi(-s)}{(-s + m^2 + \text{Re } \Pi(-s))^2 + (\text{Im } \Pi(-s))^2}. \quad (1.64)$$

Now, let us compute the vertex correction for the ϕ^3 theory. In the previous computation we took a slightly more complicated root by evaluating the path integral straight away. Now, let us go back and write down the momentum space vertex Feynman rules of the theory:

$$iZ_g g. \quad (1.65)$$

We can define an exact three-point vertex $iV_3(k_1, k_2, k_3)$, where $k_1 + k_2 + k_3 = 0$. To one-loop correction, the exact three-point is given by

$$iV_3(k_1, k_2, k_3)/g = iZ_g + i(ig)^2 \int \frac{d^d l}{(2\pi)^d} \tilde{D}_F(l^2) \tilde{D}_F((l - k_1)^2) \tilde{D}_F((l + k_2)^2) + \mathcal{O}(g^5), \quad (1.66)$$

where we have used the fact that $Z_g = 1 + \mathcal{O}(g^2)$. We shall impose the natural renormalization condition,

$$iV_3(0, 0, 0) = 0, \quad (1.67)$$

to fix the value of Z_g . Just like before, we use the Feynman integration formula to get

$$\begin{aligned} & \tilde{D}_F(l^2) \tilde{D}_F((l - k_1)^2) \tilde{D}_F((l + k_2)^2) \\ &= i^3 \int dF_3 \frac{1}{[x_1(l^2 + m^2) + x_2((l - k_1)^2 + m^2) + x_3((l + k_2)^2 + m^2)]^3} \\ &= -i \int dF_3 \frac{1}{[l^2 + m^2 - 2x_2 k_1 l + 2x_3 k_2 l + x_2 k_1^2 + x_3 k_2^2]^3} \\ &= -i \int dF_3 \frac{1}{[(l - x_2 k_1 + x_3 k_2)^2 + m^2 + x_2 k_1^2 + x_3 k_2^2]^3} \\ &= -i \int dF_3 \frac{1}{(q^2 + D)^3}, \end{aligned} \quad (1.68)$$

where $q \equiv l - x_2 k_1 + x_3 k_2$ and $D \equiv m^2 + x_2 k_1^2 + x_3 k_2^2$. Then, we have

$$\int \frac{d^d l}{(2\pi)^d} \tilde{D}_F(l^2) \tilde{D}_F((l - k_1)^2) \tilde{D}_F((l + k_2)^2) = -i \int dF_3 \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + D)^3}. \quad (1.69)$$

Perform a Wick rotation, we find

$$\begin{aligned} -i \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + D)^3} &= \int \frac{d^d q_E}{(2\pi)^d} \frac{1}{(q_E^2 + D)^3} \\ &= \frac{\Gamma(3 - d/2)}{2(4\pi)^{d/2}} D^{-3+d/2}. \end{aligned} \quad (1.70)$$

Now, substitute $\varepsilon = 6 - d$ and redefine the coupling constant just as before. We look for the asymptotic series of $\varepsilon \rightarrow 0$. That is

$$\begin{aligned}
 -i(i^3 g^2) \tilde{\mu}^\varepsilon \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + D)^3} &= \frac{ig^2}{2(4\pi)^3} \Gamma(\varepsilon/2) \left(\frac{4\pi \tilde{\mu}^2}{D} \right)^{\varepsilon/2} \\
 &\sim \frac{ig^2}{2(4\pi)^3} (2/\varepsilon - \gamma + \dots) \left(1 + \frac{\varepsilon}{2} \ln \left(\frac{4\pi \tilde{\mu}^2}{D} \right) + \dots \right) \\
 &= \frac{ig^2}{2(4\pi)^3} \left(\frac{2}{\varepsilon} - \gamma + \ln \left(\frac{4\pi \tilde{\mu}^2}{D} \right) + \dots \right) \\
 &= \frac{i\alpha}{2} \left(\frac{2}{\varepsilon} + \ln \left(\frac{\mu^2}{D} \right) + \dots \right),
 \end{aligned} \tag{1.71}$$

where we have used $\mu^2 = 4\pi e^{-\gamma} \tilde{\mu}^2$ and $\alpha \equiv g^2/(4\pi)^3$. Finally, we find the three-point vertex to be

$$iV_3(k_1, k_2, k_3)/g \sim iZ_g + i\frac{\alpha}{2} \left(\frac{2}{\varepsilon} + \int dF_3 \ln(\mu^2/D) + \dots \right) + \mathcal{O}(g^5). \tag{1.72}$$

By setting $Z_g = 1 + C$, we find

$$V_3(k_1, k_2, k_3)/g \sim 1 + \alpha \left[\frac{1}{\varepsilon} + \ln(\mu/m) + \frac{1}{2} \int dF_3 \ln(m^2/D) + \dots \right] + C + \mathcal{O}(\alpha^2). \tag{1.73}$$

Again, we let C to take the form

$$C = -\alpha \left[\frac{1}{\varepsilon} + \ln(\mu/m) + \kappa_C \right] + \mathcal{O}(\alpha^2), \tag{1.74}$$

which gives

$$V_3(k_1, k_2, k_3)/g \sim 1 - \frac{\alpha}{2} \int dF_3 \ln(D/m^2) - \alpha \kappa_C + \mathcal{O}(\alpha^2). \tag{1.75}$$

Now, we impose the renormalization condition, $V_3(0, 0, 0) = 0$, which gives $D = m^2$, and hence

$$\kappa_C = 0. \tag{1.76}$$

Therefore, our final expression for the three-point vertex to first order in α is given by

$$V_3(k_1, k_2, k_3)/g \sim 1 - \frac{\alpha}{2} \int dF_3 \ln(D/m^2) + \mathcal{O}(\alpha^2). \tag{1.77}$$

Now, we have determined all three of the renormalization factor to one-loop order.